Fiscal Year:	FY 2008	**Task Last Updated:**	FY 06/18/2008
PI Name:	Pandya, Abhilash Ph.D.		
Project Title:	Real-Time Augmented Reality Development and Human Factors Assessment for the Special Purpose Dexterous Manipulator		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	07/07/2004	**End Date:**	03/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 3/31/2008 per PI (9/07)		
Key Personnel Changes/Previous PI:	Keshav Chintamani (Ph.D Candidate) Thomas Overgaard (B.S. Student)		
COI Name (Institution):	Baker, Kenneth (NASA/JSC) Maida, James (NASA/JSC) Ellis, Darin (Wayne State University) Chien, Chiun-Hong (NASA/JSC/HEI) Cao, Alex (Wayne State University) Chintamani, Keshav (Wayne State University)		
Grant/Contract No.:	NNJ04HI11A		
Performance Goal No.:			
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Task Description:	The major goal of this Rapid Technology Development grant is to provide the operator of the Special Purpose Dexterous Manipulator (SPDM) with simple, intuitive cues for control of the manipulator during grasping, removing, stowing and inserting orbital replacement units (ORUs) as opposed to just looking at multiple video views which requires extensive training and can be error prone. The science of presenting and displaying complex 3D images in an augmented and operationally meaningful way is the basis of this Space Human Factors Engineering research proposal. The technical development will be validated and proven with the appropriate Human Factors testing. Efficient methods of remote image data display will be implemented and tested against more traditional methods. The proposal's objective is to improve the human-robot interface to reduce operator work load, task time and to simplify operator training to make skill-based training practical as opposed to task-based training with particular focus on the SPDM and its training facility the DMT. Because of the risky environment of space, teleoperations are commonplace for NASA. Visualization of the remote data and its human performance impact are important to succesful teleoperations. During space missions, astronaut robotic operators primarily depend on visual feedback that is provided by strategically placed video cameras. Operating a robotic arm from indirect visual information poses many technical and human performance challenges for the operator. Astronauts performing remote functions like operating the SPDM or the RMS need to maintain mental 3D real-time environment maps for successful operations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The earth significance of this proposal is that successful completion of our first specific aim for this proposal (the core Augmented Reality Specifications) will provide an easily translatable kinematics-based AR system specifications for medical robotics. In addition, several important features in Medical Robotics systems (voice recognition and control, motion scaling and built-in structural intelligence) can also be of interest to NASA robotics. Medical robotics is an up-and-coming technology with tremendous potential and has some of the same visualization difficulties facing astronaut robotic operators. Medical Robotic systems like the Zeus (Computer Motions Inc.) and daVinci (Intuitive Inc.) (both presently being used in our collaborating sister facilities; Henry Ford Hospital—Zeus system and Children's Hospital—daVinci System both in Detroit, Michigan) are master-slave systems for minimally invasive surgery. Direct linkage of medical robotic systems to patient data and the optimal visualization of that data for the surgical team are of critical importance for successful operations.
Task Progress:	There are three areas of progress: (1) Physically based Augmented Reality for Remote Robot Teleoperation Applications in Training and Simulation Space station service telerobots have been extremely useful to the space program and the astronaut. However, the high cost of using these robots in pre-mission astronaut training calls for low cost and efficient training platforms. In addition, local gravitational fields (i.e. microgravity) cannot be accurately simulated in robot tele-operation training platforms on Earth. We have developed a robot tele-operation interface that allows remote camera views to contain dynamic simulations using Augmented Reality (AR). In these views, virtual and physical robot arms co-operate in the remote worksite and can be manually (end-effector) controlled. In the current system, a virtual payload is placed in the remote worksite views and operators can control the virtual robot's end-effector to grasp and transport this payload in any user defined gravity condition. The physical effects arising from these interactions and the resulting motion of the payload when a collision with the robot occurs is simulated. There are practical implications of such simulations, particularly in pre-mission training of astronauts. We have presented and demonstrated this work to NASA/JSC (Astronaut Nancy Currie) and to the head of Space Station Robotics Shakeel Razvi. (2) Improved Tele-manipulator Navigation with Augmented Coordinate Frames The application of simple Augmented Reality (AR) cues for remote robot tele-operation has been implemented and tested with human participants. Space robot arms, for example, are manipulated by orienting and translating the body fixed coordinates at the end-effector (EE) using hand controllers with cameras providing visual feedback. The pose of the end-effector seen on 2-dimensional images however, challenges operator performance due to misalignments between the displayed EE axes and the hand controller (HC) axes. In this study, the coordinate system of the EE is graphically overlaid in 3-dimensions on the live video with unique colored axes using AR. The same color scheme is used to label the appropriate axis on the HCs. Operators can use these color cues to map each axis on the HCs to the corresponding colored axis on augmented coordinates at the EE to obtain EE motion in the desired direction. A within-participant experiment comparing the EE trajectory distance, deviation from path and navigation errors was used to determine the effectiveness of augmented coordinates over conventional tele-operation without augmented coordinates. Significant reductions in EE trajectory distance, deviation from path and navigation errors were observed when participants manipulated the remote robot with augmented coordinates. Observations and inferences demonstrate that the use of AR in remote robot arm tele-operation is beneficial to the operator. (3) Operator Performance in Robotic Tele-operation Using End-effector vs. Disjointed Wrist Kinematics Control Tele-operated robot arms in space are controlled by manipulating points of resolution (POR) at the end-effector, wrist and other joints using joystick like hand controllers. Depending on motion constraints at the POR, the POR's motion perceived by the operator can vary. Typical tele-operation is performed by rotation and translation of the end-effector (EE-POR). Incongruence between the EE-POR axes and joystick axes due to misaligned exocentric camera views along with prolonged closed loop operation can lead to high levels of mental workload and longer task times. Control at the wrist POR (WR-POR) is proposed as a means to ease the workload levels and improve operator performance. Twelve participants performed payload insertion tasks using a remote robot arm equipped with the EE-POR and WR-POR modes. Subjective mental workload scales (NASA-TLX) were obtained along with task time, distance and reversal error data. Time and subjective workload were found to be lower with WR-POR, although average path distances were longer with reversal errors indistinguishable for both modes. Findings confirm that it is possible to ease the inherent difficulties in remote tele-operation by introducing an alternative POR control scheme like WR-POR. Although this POR control may not be dexterous enough for precise tasks, its use in conjunction with EE-POR control can ease operator workload. This work can be applicable to any tele-operated robotic system (space, medical and military) where the user has to control multiple degrees of freedom at the POR.

Bibliography Type:	Description: (Last Updated: 06/30/2008)
Papers from Meeting Proceedings	Chintamani K, Overgaard T,Tan C-A, Ellis RD, Pandya AK. "Physically-based Augmented Reality for Remote Robot Teleoperation." Institute of Industrial Engineers Research Conference, Vancouver, B.C., Canada, May 17-21, 2008. IIE Annual Conference and Expo 2008. , May-2008
Papers from Meeting Proceedings	Ellis RD, Pandya A, Chintamani K, Awasthi A, Nawab A. "A Test-Bed for Evaluation of Augmented Tele-operation." Institute of Industrial Engineers Research Conference, Nashville, TN, May 19-23,2007. IIE Annual Conference and Expo 2007. , May-2007
Papers from Meeting Proceedings	Nawab A, Chintamani K, Ellis RD, Pandya AK. "Joystick mapped Augmented Reality Cues for End-Effector controlled Teleoperated Robots." IEEE Virtual Reality Conference 2007, Charlotte, NC, USA, March 10-14 2007. Proceedings of the IEEE Virtual Reality Conference, p. 263-266, 2007. http://dx.doi.org/10.1109/VR.2007.352496 , Mar-2007
Papers from Meeting Proceedings	Chintamani K, Nawab A, Cao A, Ellis RD, Auner G, Pandya AK. "Comparing Two Kinematics Methods For Telerobotic Control Applications." 50th Annual meeting of the Human Factors and Ergonomics Society, San Francisco CA, October 2006. Human Factors and Ergonomics Society Annual Meeting Proceedings, Aerospace Systems volume, p. 126-130(5), 2006. , Oct-2006